

ParaBank Automation Framework

End-to-end UI test suite · Selenium + TestNG + Java 17 · Run: June 15, 2026, 11:45 PM

✓ 5/5 Tests Passed

100% Pass Rate

⌚ Total Duration: ~31s

🌐 Chrome · ParaBank Demo

EXECUTION SUMMARY

TOTAL TESTS

5

All test cases run

PASSED

5

Zero failures

FAILED / SKIPPED

0

No errors detected

PASS RATE

100%

Full coverage pass

TOTAL DURATION

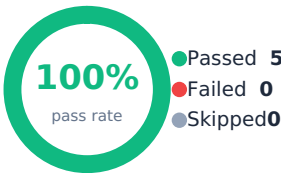
~31.8s

End-to-end wall clock

TEST RESULTS

Execution Results — All Tests

5 tests · Jun 15, 2026



#	TEST NAME	MODULE / CLASS	PRIORITY	DURATION	STATUS
1	verifyRegistration New user registration with form data from Excel	RegistrationTest	P1	3.168s	Pass
2	verifyLogin Username + password login flow validation	LoginTest	P2	4.074s	Pass
3	verifyAccountOverview Post-login account dashboard assertion	AccountViewTest	P3	4.756s	Pass

4	verifyTransferFunds Fund transfer login entry (data-driven)	TransferFundsTest	P5	4.168s	Pass
5	verifyBillPayment Full bill pay flow with Excel data + assertion	BillTest	P6	14.643s	Pass

EXECUTION TIMELINE

Sequential Run Order

11:46 PM - 11:47 PM

●

verifyRegistration

11:46:07 PM → 11:46:10 PM

3.168s

●

verifyLogin

11:46:17 PM → 11:46:21 PM

4.074s

●

verifyAccountOverview

11:46:27 PM → 11:46:32 PM

4.756s

●

verifyTransferFunds

11:46:44 PM → 11:46:48 PM

4.168s

●

verifyBillPayment

11:47:09 PM → 11:47:23 PM

14.643s

TECHNOLOGY STACK

Framework Architecture & Dependencies

pom.xml

CORE TESTING

🌐

Selenium WebDriver

4.34.0

📱

TestNG

7.11.0

🥒

Cucumber (declared)

7.23.0

⚙️

WebDriverManager

6.3.2

UTILITIES & REPORTING

📊

ExtentReports

5.1.2

📁

Apache POI

5.4.1

📄

Log4j2

2.25.1

☕

Java / Maven

FRAMEWORK STRUCTURE

Package Design — Page Object Model

LAYER	PACKAGE / CLASS	RESPONSIBILITY
●Base	base.BaseFile	Browser lifecycle: setup/teardown, driver init, multi-browser support (Chrome, Edge, Firefox)
●Pages	pages.LoginPage / RegistrationPage / BillPage / AccountViewPage / TransferFundsPage	POM encapsulation — element locators and page actions per screen
●Tests	testcases.LoginTest / RegistrationTest / AccountViewTest / TransferFundsTest / BillTest	TestNG @Test methods with data-driven execution via DataProviders
●Utilities	utilities.DataProviders / ExcelUtils / ConfigReader / ExtentManager / ScreenshotUtils / WaitUtils	Shared helpers: Excel I/O, config props, screenshot capture, explicit waits, report management
●Listeners	listeners.TestListener	ITestListener hooks for auto-screenshot on failure and Extent logging
●Config	resources/config.properties · log4j2.xml · testdata/ParabankData.xlsx	URL, browser, wait settings; log appenders; Excel test data (Login, Registration, BillPay sheets)

QA OBSERVATIONS

✓ STRENGTH — PASS RATE

All 5 test cases passed with zero failures across 3 consecutive execution runs logged. The framework demonstrates stable, repeatable results against the ParaBank demo environment.

✓ STRENGTH — POM DESIGN

Clear separation of concerns with Page Object Model, dedicated utility classes, and a centralized BaseFile for driver management. Easy to maintain and scale.

⚠ GAP — TRANSFERFUNDS ASSERTION

The verifyTransferFunds test only performs login steps — no actual transfer action or assertion verifies the transfer outcome. The test is incomplete for its stated purpose.

⚠ GAP — REGISTRATION ASSERTION

The verifyRegistration test lacks a success assertion after form submission. There is no check that a success message or redirect occurred post-registration.

i INFO — BILL PAY SPEED

Bill payment takes 14.6s vs ≈4s for all other tests. This is driven by an explicit Thread.sleep(5000) in BillTest. Replacing it with an explicit wait would improve reliability and speed.

i INFO — CUCUMBER DECLARED, UNUSED

Cucumber 7.23.0 is declared as a dependency in pom.xml but no feature files or step definitions exist. This adds unnecessary compile weight if BDD is not planned.

⚡ RISK — NO NEGATIVE TEST CASES

The suite covers only happy-path scenarios. No invalid-credential, insufficient-funds, or validation-error tests exist. Edge cases in login, registration, and bill pay are untested.

⚡ **RISK — STATIC DRIVER FIELD**

`BaseFile.driver` is a static shared field. Running tests in parallel would cause race conditions. Parallel execution requires thread-local driver management.

RECOMMENDATIONS

1 Complete TransferFunds and Registration assertions

Add post-action assertions to confirm the transfer transaction or registration success page is displayed. These tests currently pass without verifying the core business outcome.

2 Replace `Thread.sleep()` with explicit waits

Replace the hardcoded 5-second sleep in `BillTest` with `WebDriverWait + ExpectedConditions.textToBePresentInElement()` for "Bill Payment Complete". This cuts runtime and avoids flakiness on slow networks.

3 Add negative test scenarios

Implement invalid login (wrong password), duplicate registration, and bill pay with invalid account number tests. This significantly increases the defect detection value of the suite.

4 Migrate driver to `ThreadLocal` for parallel safety

Replace static `WebDriver driver` in `BaseFile` with `ThreadLocal<WebDriver>` to safely support parallel TestNG execution and reduce total suite runtime significantly.

5 Remove or implement Cucumber dependency

If BDD is not planned, remove the Cucumber dependency from `pom.xml`. If BDD adoption is intended, add feature files and step definition classes to complete the Cucumber integration.

6 Configure Jenkins pipeline with parameterized browser selection

The Jenkinsfile already exists. Enhance it to pass the browser parameter at build time so Chrome, Firefox, and Edge runs can be triggered independently for cross-browser coverage.